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ScienceDirect
Journal of Non-Crystalline Solids 354 (2008) 1128–1132
JOURNAL OF NON-CRYSTALLINE SOLIDS
www.elsevier.com/locate/jnoncrysol

Dielectric properties and crystalline characteristics of borosilicate glasses

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Available online 26 November 2007

Abstract

Packaging substrates play a significant role in the development of large-scale integrated circuit. Low dielectric constant materials are expected in order to reduce the time delay of signal propaganda. On the other hand, sintering aid is required for low temperature co-firing process. Glass/ceramic composite system is considered one of the most promising candidates because of its both low dielectric constant and low temperature co-firable properties. Borosilicate glass is an important component of the substrate material as a sintering agent. To reduce the firing temperature, some of the alkali metal or bivalent metal oxides must be introduced, often at the expense of dielectric properties as well as crystallization behavior. The paper has designed two groups of glasses doped with different metal oxides. Dielectric constant and loss, crystalline characteristics as well as softening point and thermal expansion coefficient of these glasses have been studied. The glass system is characterized with lower dielectric constant (5–6) and loss (10^{-2}–10^{-3}), higher electric resistivity (10^{12-13} Ω cm). Softening point of the glass is also satisfactory for low temperature co-firing process. Furthermore, the glass shows a severe tendency of phase separation and followed by crystallization, with cristobalite as the main crystal.

PACS: 61.43.Fs; 64.75.+g; 77.22.−d

Keywords: Crystallization; Dielectric properties, relaxation, electric modulus; Borosilicates

1. Introduction

Packaging substrates play a significant role in the development of information technology. Technical demand for large-scale integrated (LSI) circuit is increasing. Signal propaganda delay is proportional to the square root of dielectric constant of LSI circuit packaging substrates. Low dielectric constant materials are preferred to reduce the time delay, especially for high frequency application. On the other hand, low temperature sintering agents are required for low temperature co-firing (LTCF) process. Glass/ceramic composite system is considered one of the most promising candidates because of its both low dielectric constant and low temperature co-firable properties [1–3].

Borosilicate glass is a typical glass component in glass/ceramic composites for its excellent dielectric properties. Doped with various oxides, although as less as possible and often at expense of dielectric properties, the glass is characterized with low softening temperature, which is assumed a decisive factor to the sintering temperature of the composites. On the other hand, the glass is apt to separate into two phases, silica rich and boric oxide rich phases, and lead to cristobalite precipitation during the firing process, deteriorating the sinterability of glass/ceramic composites and the properties of packaging substrates as well [4,5]. In the present work, two groups of glasses doped with a variety of modifying cations are melted and their thermal properties, dielectric properties and crystalline characteristics are investigated. The objective of the study is to reduce the firing temperatures by using glass as sintering aid without sacrificing the dielectric properties.

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doi:10.1016/j.jnoncrysol.2007.01.099

Table 1
Compositions of glasses in Group I (mol%)

Sample no.	SiO_2	B_2O_3	Al_2O_3	CaO	Li_2O	Na_2O	K_2O
11	62	25	1.2	5.4	6.4	–	–
12	62	25	1.2	5.4	–	6.4	–
13	62	25	1.2	5.4	–	–	6.4

Table 2
Compositions of glasses in Group II (mol%)

Sample no.	SiO_2	B_2O_3	Al_2O_3	CaO	MgO	ZnO	Na_2O	Li_2O
21	62	25	1.2	5.4	–	–	3.2	3.2
22	62	25	1.2	–	5.4	–	3.2	3.2
23	62	25	1.2	–	–	5.4	3.2	3.2

2. Experimental methods

Two groups of low dielectric constant glasses were prepared and oxide compositions are listed in Tables 1 and 2. Various alkali oxides were introduced in Group I with fixed amount of SiO_2, Al_2O_3, B_2O_3, CaO. Besides, different bivalent metal oxides were added into the Group II while the amount of SiO_2, Al_2O_3, B_2O_3, Na_2O and Li_2O remain unchanged. The raw materials used for glass batches were SiO_2, $B(OH)_3$, $Al(OH)_3$, $CaCO_3$, $MgCO_3$, ZnO, Na_2CO_3, K_2CO_3, Li_2CO_3 as pure chemical reagents. Glass batches were melted and refined in a platinum crucible for 6 h at 1480 °C in an electric heating furnace. The glass melts were cast into a stainless steel mould to form glass slabs. Then, the glasses were annealed at 550 °C in an annealing oven for 1 h and cooled down slowly to room temperature.

Samples for thermal expansion coefficient (TEC) and glass transition temperature were cut and ground in the form of bar of 50 mm in length and 3 mm in diameter. TECs were measured by using fused silica dilate meter at heating rate 4 °C/min. 0.65 mm in diameter and 235 mm long glass fibres were drawn for the softening point measurement by using Littleton softening point meter at the heating rate of 4 °C /min. For liquidus temperature, phase separation and crystallization temperatures range, test specimens were prepared by crushing the glasses into small pieces and filled in a tray for heat treatment in a gradient temperature oven for 2 h. As to the study of crystallization behaviour, glass samples were firstly ground into powder and then treated at 900 °C for 20 h for nuclei formation and crystal growth. Crystal precipitations were identified by X-ray diffraction analyser. Samples for electric properties measurement were prepared in the form of discs of 25 mm in diameter and 2 mm in thickness. Dielectric properties and electric resistivity were measured by dielectric spectrometer and resistivity meter, respectively.

Table 3
Transition temperatures, softening points and thermal expansion coefficients of the glasses

Sample no.	T_g (°C)	Softening point, T_f (°C)	TEC ($\times10^{-7}$/°C) (25–400 °C)
11	495	Not available	54.97
12	525	693	60.64
13	540	705	67.35
21	500	717	57.43
22	480	712	54.79
23	460	706	51.71

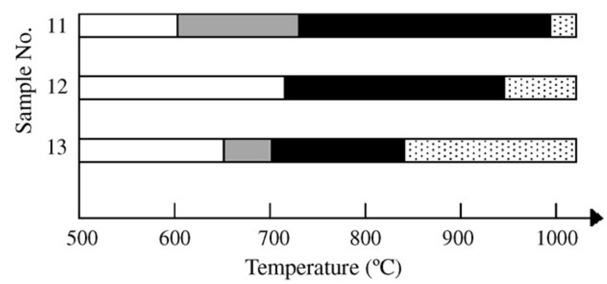

Fig. 1. Results of gradient temperature treatment of glasses doped with alkali oxides in Group I for 2 h. The white area represents the samples without any change, the grey area phase separation, the dark area crystallization, and dotted area glass fusion. All lines were marked with the help of microscope.

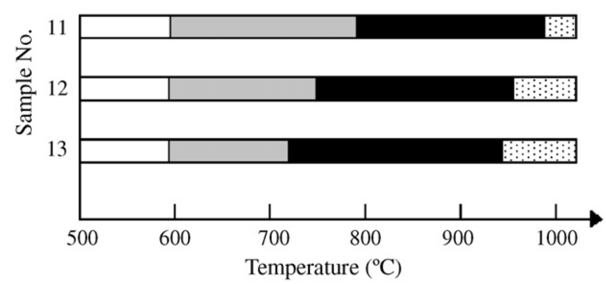

Fig. 2. Results of gradient temperature treatment of glasses doped with bivalent oxides in Group II for 2 h. The white area represents the samples without any change, the grey area phase separation, the dark area crystallization, and dotted area glass fusion. All lines were marked with the help of microscope.

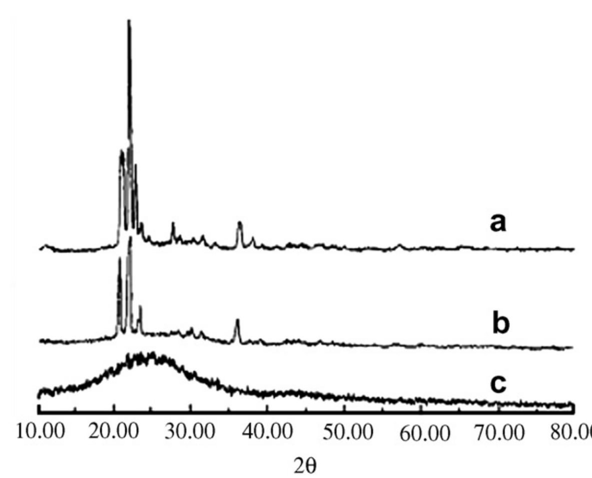

Fig. 3. X-ray diffraction analyses of glasses in Group I, scanning speed 8°/min, incident light wavelength 1.54 Å with copper target, tube 40 kV/ 60 mA: (a) Sample-11, (b) Sample-12, (c) Sample-13.

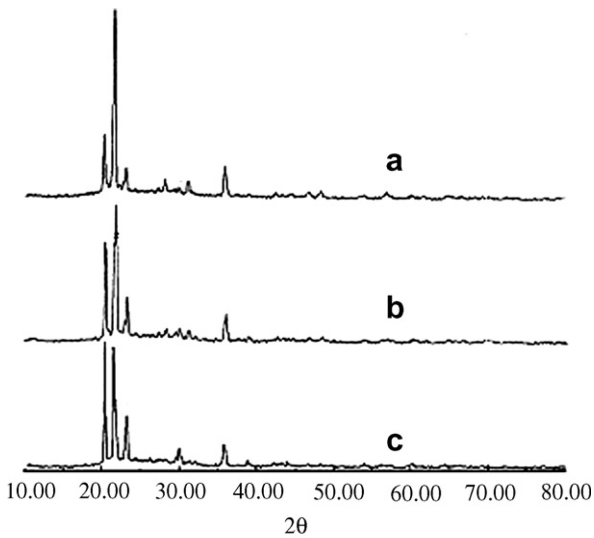

Fig. 4. X-ray diffraction analyses of glasses in Group II, scanning speed 8°/min, incident light wavelength 1.54 Å with copper target, tube 40 kV/60 mA: (a) Sample-21, (b) Sample-22, (c) Sample-23.

Table 4
Dielectric properties and resistivities of the glasses

Sample no.	Dielectric constant (10 MHz)	Dielectric loss (10 MHz) $\times 10^{-2}$	Resistivity (Ω cm)
11	5.58	4.97	0.73×10^{13}
12	6.03	4.50	0.66×10^{13}
13	6.05	4.41	1.17×10^{13}
21	5.52	1.82	2.70×10^{13}
22	6.11	4.34	1.02×10^{13}
23	6.03	4.43	0.30×10^{13}

3. Results

Glass transition temperatures (T_g), softening points (T_f) and TECs are summarized in Table 3. Transition temperatures of the glasses were deduced from the expansion curves. Results of gradient temperature treatment of the glasses in Group I and Group II are illustrated in Figs. 1 and 2, respectively. X-ray diffraction patterns of the glasses in Group I and Group II heat-treated at 900 °C for 20 h are given in Figs. 3 and 4 separately. Dielectric properties and resistivities of the glasses are listed in Table 4.

4. Discussions

4.1. Transition temperatures, softening points and thermal expansion coefficients

For glass/ceramic composites, sintering take place at the presence of liquid phase and with the help of viscous flow of the glass phase. Softening point of the glass is considered one of the key factors for the firing temperature of the composites. When the glasses are doped with alkali ions, lithium ion in Sample-11 depolarises Si–O tetrahedra by its high electrostatic field strength. It reduces melting temperature by decreasing the high temperature viscosity of the glass melt, whereas in the present work, softening point of sample-11 is not available because of the difficulty of fibre sample preparation result from its severe phase separation immediately after casting. However, we can deduce that it must have the lowest firing temperature from the fact that T_g of the Sample-11 is the lowest among that of all the glasses in Group I. Comparing with potassium oxide, sodium oxide is a more effective flux as listed in Table 3 that softening temperature of Sample-12 is lower than that of Sample-13. In comparison between the glasses in Group II, the lowest softening point of Sample-23 containing ZnO can be explained by the polarization of Zn^{2+}, which results in less bonding force of Si–O bond, and followed by glass viscosity decrease.

Packaging substrates are presumably to have the expansion matching that of silicon (34×10^{-7}/°C) in order to eliminate mechanical stress. In this case, TECs of all the glasses are ranging from 50×10^{-7}/°C to 70×10^{-7}/°C. However, they are second important because TEC of the composites is expected according to the mixture rule based on volume fraction and property of original ingredients. Lower TEC of the composites substrate can be approached by introducing less expansible ceramics in glass/ceramic composites process.

4.2. Phase separation and crystalline characteristics

4.2.1. Influence of alkali oxides on the crystalline behavior

Borosilicate glass exhibits severe tendency of phase separation, which subsequently promotes crystallization as phase separation provides the material with phase boundary and reduces critical nucleation energy. The influence of the alkali ions on the phase separation is related to the bonding force between alkali ions and O^{2-}. Fig. 5 [6] illustrates R_2O–B_2O_3–SiO_2 ternary phase diagrams. Obviously, Li_2O–B_2O_3–SiO_2 system represents the most intense phase separation tendency and has the largest phase separation area due to the highest electrostatic field strength of lithium ion. As shown in Fig. 1, liquidus temperature lists in the order of Sample-11 > 12 > 13. Crystallization temperatures of the glasses are ranging from 730–980 °C, 720–940 °C and 695–840 °C, respectively for Sample-11, 12 and 13. These results fully agree with the phase diagrams and suggest that the negative effect of the lithium oxide must be taken into consideration.

The X-ray diffraction analysis in Fig. 2 has identified that except for Sample-13, the crystal precipitated during the heat treatment for Sample-11 and 12 is cristobalite. Since the coefficient of expansion of cristobalite is high, TECs of the composites with the presence of cristobalite will be much higher than the value expected. It also provides us with the fact that potassium oxide containing Sample-13 has less crystalline tendency. No crystal can be detected and the glass remains a perfect vitreous state even

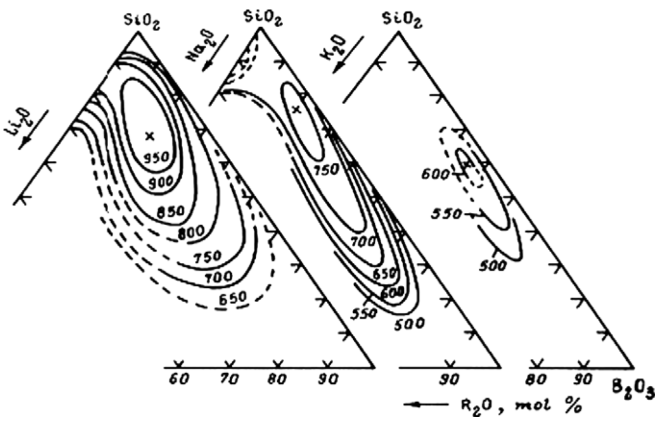

Fig. 5. R_2O–B_2O_3–SiO_2 ternary phase diagram.

though a long time at 900 °C heat treatment was performed.

4.2.2. Influence of bivalent oxides on the crystalline behavior

Comparing with R_2O, RO in glass is more capable to promote phase separation. The higher electrostatic field strength of the bivalent metal ions is responsible for the occurrence of phase separation. As illustrated in Fig. 6 [6], SiO_2–B_2O_3–PbO and SiO_2–B_2O_3–BaO systems have a smaller phase separation area. However, PbO or BaO are not introduced since these cations are apt to polarize, which will increase the dielectric constant of the glass. CaO–B_2O_3–SiO_2 in the ternary phase diagram appears smaller phase separation area. Refer to the ion and covalent bond radius of Ca^{2+}, Mg^{2+}, and Zn^{2+} listed in Table 5 [7], Ca^{2+} is bigger in its size and lower in electrostatic field strength compare with Mg^{2+}, therefore, smaller in phase separation area. Zn^{2+} is bigger in its ion radius but smaller covalent bond radius. The polarization of 18 electron layer structure results in higher strength of covalent bond to attract O^{2-}, and thus larger in phase separation area. In our study, CaO, MgO and ZnO are chosen as the dopants. Temperatures of the crystallization of the glasses are ranging from 780–980 °C, 740–960 °C and 720–950 °C, respectively for

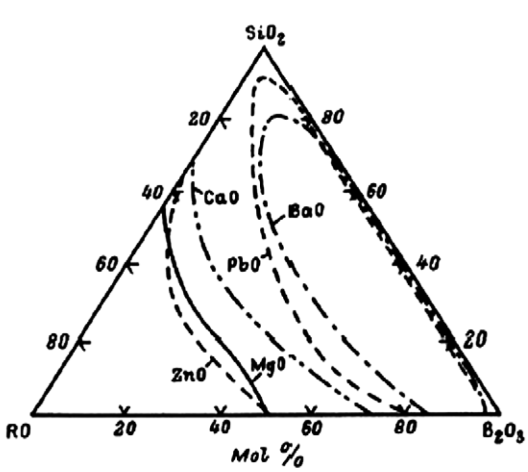

Fig. 6. RO–B_2O_3–SiO_2 ternary phase diagram.

Table 5
Ion and covalent bond radius of Ca^{2+}, Mg^{2+}, Zn^{2+}

	Ca^{2+}	Mg^{2+}	Zn^{2+}
Ion radius (Å)	0.99	0.65	0.74
Covalent bond radius (Å)	1.74	1.36	1.25

Sample-21, 22 and 23. Liquidus temperatures of the glasses are in the order of Sample-21 > 22 > 23. Crystal precipitated during the heat treatment for all of these three glasses is the same as the previous group, cristobalite, as identified by X-ray diffraction analysis in Fig. 4.

4.3. Electric properties

4.3.1. Dielectric constant

The dielectric constant of a glass results from electronic, ionic, and dipole orientation contributions to the polarizability. It increases with the increase of total polarization rate. Only electronic contribution (α_e) and ionic contribution (α_i) are taken into consideration under the high frequency condition. The α_e and α_i are related to ion radius and atomic weight, respectively. The larger radius of the ion results in the higher α_e. The higher atomic weight results in the lower α_i. The lowest dielectric constant of Sample-11 in Group I can be explained by the minimum radius of lithium ion. The similarity of the dielectric constant of Sample-12 and 13 is attributed to the similarity of the total polarizability contributed by Na^+ and K^+. In another word, K^+ is larger in size and contributes higher α_e than Na^+, while Na^+ is lower in atomic weight and contributes higher α_i than K^+.

Based on the experiment results, atomic weight of the bivalent ions in Group II plays a key role for the dielectric constant. Comparing with Mg^{2+}, Ca^{2+} is higher in its atomic weight and contributes smaller α_i, therefore, a smaller dielectric constant. As for Sample-23, although Zn^{2+} is higher in its atomic weight, the higher electron polarization α_e result from the 18 electron structure explains the similarity dielectric constant between Sample-22 and Sample-23.

4.3.2. Dielectric loss

Energy losses in dielectrics result from three primary processes, they are: ion migration losses, ion vibration and deformation losses, and electron polarization losses. The major factor affecting the use of ceramic materials is the ion migration losses [8]. Modifying cations have a significant effect on the dielectric loss of the glasses. The order of mobility of alkali ions is $Li^+ > Na^+ > K^+$, the dielectric losses in Group I follow in the same order. If the same bivalent metal oxide is introduced, dielectric losses are expected lower due to the mixed alkali effect, which explains the difference between Sample-21 and all the glasses in Group I. In Group II, the higher loss of Sample-22 comparing with that of Sample-21 is attributed to the higher mobility of Mg^{2+}, whereas for Sample-23, most probably, to the polarization nature of Zn^{2+}.

4.3.3. Electrical conductivity

In glasses containing alkali oxides, the current is carried almost entirely by alkali ions. The mobility of these ions is much larger than that of the bivalent oxides. The conduction characteristics are determined by the concentration and mobility of the alkali ions. In our study, the electrical conductivity are mostly depends on the mobility of the alkali due to the same concentrations of alkali. The mobility of the ions is related to the size of the ions, the bonding force between R^+ and O^{2-}, and the strength of network. The highest electrical conductivity of Sample-12 in Group I can be explained that Na^+ is smaller than K^+ in its size and lower than Li^+ in its electrostatic field strength. The lowest resistivity of Sample-23 in Group II probably is attributed to the polarization of Zn^{2+}, and consequently diminished the $[SiO_4]$ network strength.

5. Conclusions

Two groups of borosilicate glasses doped with alkali metal oxides and bivalent metal oxides are prepared. The glasses are characterized with low dielectric constant and loss, higher electric resistivity, and low temperature firing properties. On the other hand, the glasses have a strong tendency of phase separation and followed by crystallization. Cristobalite is precipitated during heat treatment, which is detrimental and must be suppressed in multi-layer glass/ceramic composites process.

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